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Microwave-Assisted Solvent Extraction

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Introduction

From simply heating up a meal to preparing a sample for analysis via a multistep process, microwave technology finds various uses in both homes and analytical laboratories. It can help reduce the time taken to complete many processes, and so it is very useful with many types of samples. How microwaves speed up some processes is not fully understood, however. The main factors to be considered here include sample transport properties, molecular agitation, the heating of solvents above their boiling points, and, in some cases, product selectivity.

Analytical chemists began using microwaves in 1975 to digest biological samples. They used domestic microwave ovens then to heat a mixture of a sample and digestion acids rapidly to its atmospheric boiling point in an Erlenmeyer flask. Since this, microwave-assisted methods have been used widely in analytical chemistry. The latest advances in the use of microwaves in various chemical fields include sample digestion for elemental analysis, solvent extraction, sample drying, moisture measurement, analyte adsorption and desorption, cleaning up samples, chromogenic reactions, solid-phase retention, elution, distillation, microwave plasma atomic spectrometry, and synthetic reactions, among others.

Microwave-assisted extraction (MAE) is the process by which microwave energy is used to heat solvents in contact with samples (mainly solid samples) and to partition compounds of interest from the sample into an extractant. Extractions are done in closed or open microwave-transparent vessels where a solvent and sample are combined and then exposed uniformly to microwave energy. Partitioning may occur by any one or several of the following three heating mechanisms: (1) a single solvent or mixture of solvent possessing high dielectric loss coefficients; (2) solvent mixtures of high and low dielectric loss; and (3) susceptible samples with a high dielectric loss in a solvent of low dielectric loss.

Although the traditional Soxhlet and solvent extraction techniques are widely accepted, they have inherent limitations and problems. Thus, Soxhlet extraction requires 12–24 h in most cases and uses high volumes of organic solvents (hundreds of milliliters). In contrast to conventional methods, MAE can reduce the extraction time to less than 30 min and solvent consumption to under 50 ml. Moreover, the recoveries obtained with MAE are mostly comparable with those provided by alternative methods.

Unlike digestion, MAE is used mainly with soil and sediments rather than with biological samples. As regards analytes, MAE is applied primarily to organic pollutants such as polycyclic aromatic hydrocarbons (PAHs), polychlorinated biphenyls (PCBs), and pesticides rather than to metal species, where other extraction alternatives, such as ultrasound-assisted extraction, are usually more effective.

This article describes the principal applications of microwaves to sample extraction, with special emphasis on solid samples, which have been the most used so far. The description is preceded by a discussion of the equipment used, which can be of the open or closed type, depending on whether they operate at atmospheric pressure or at a higher level and whether they use multimode or focused microwaves. Selected special laboratory designs are also commented on, as is the possibility of coupling of some of these devices to subsequent steps of the analytical process. Before applications are dealt with, the main variables affecting MAE are also examined.

Microwave-Assisted Extractors

Microwave-assisted extractors can be classified into two groups, according to the way microwave energy is applied to the sample. In multimode systems, the microwave radiation is dispersed randomly in a cavity, and so every zone in the cavity and the sample it contains is irradiated. The single-mode or focused system allows focusing of the microwave radiation on a restricted area where the sample is placed for subjection to a much stronger electromagnetic field than in the previous case. **Figure 1** shows a general scheme of both types of extractors. Usually multimode systems (**Figure 1A**) are of the closed-vessel type, where the microwave-assisted treatment is conducted at a high pressure, whereas focused systems (**Figure 1B**) are of the open-vessel type, in which microwaves are applied at atmospheric pressure. This mutual association, however, is incorrect as some devices that operate at a high pressure use focused microwaves, and domestic ovens have been used with multimode sample irradiation but at atmospheric pressure.

Closed-vessel systems have not been used so much for sample extraction as they are primarily used for sample digestion, whereas the main application of open-vessel systems has been solid sample extraction.

Both multimode and focused microwave devices consist of four major components:

1. The 'magnetron' or microwave generator, where the microwave energy is produced.
2. The waveguide, used to propagate the microwaves from the magnetron to the microwave cavity.
3. The applicator, where the sample is placed. It can be a multimode cavity where microwaves are randomly dispersed or the waveguide itself. In the latter case the sample vessel is placed directly inside to focus the microwave radiation onto the sample.
4. The circulator, which allows microwaves to pass only in the forward direction.

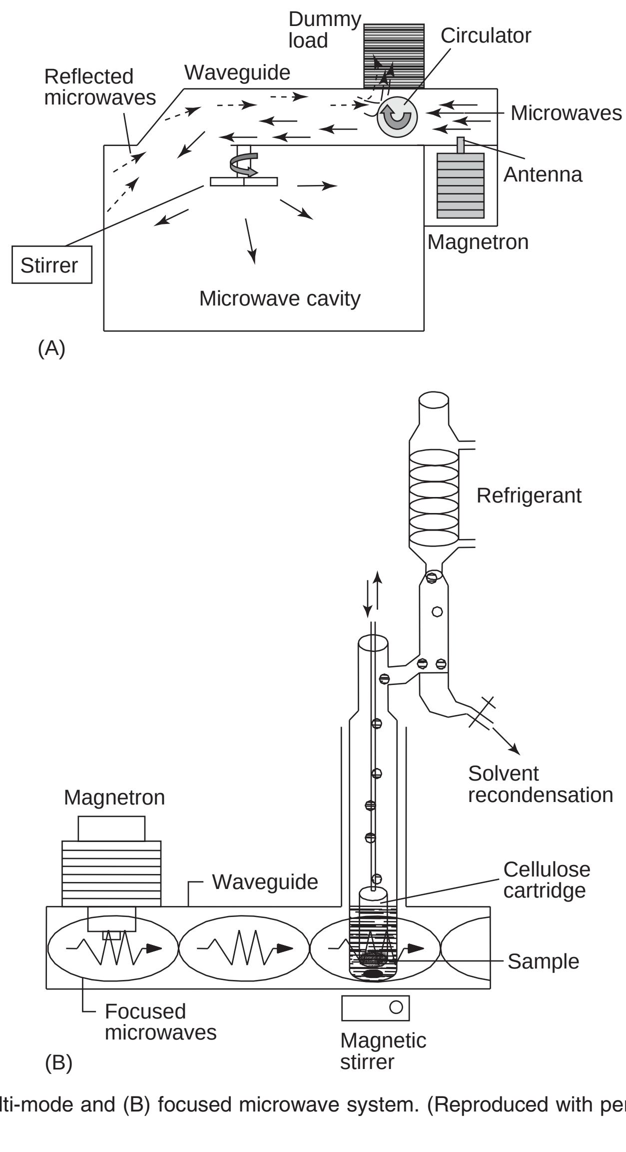

Figure 1 Scheme of a (A) multi-mode and (B) focused microwave system. (Reproduced with permission from Springer.)

Closed-Vessel Microwave Systems

The earliest microwave systems for analytical purposes were closed vessels with a multimode cavity. They usually allowed processing of several samples at the same (in a carousel) under pressure and temperature feedback control. Closed vessels exist basically in two different forms. One encompasses noninsulated, relatively thin, single-walled fluoropolymer vessels. These vessels have minimal insulating characteristics and allow large amounts of heat to escape. The other type of closed vessel is a well-insulated container, usually of very thick-walled fluoropolymer, or one with a very thick outer layer or casing (or both). These vessels retain heat very efficiently, and so they do not allow rapid cooling when ambient air is forced over them within the microwave cavity.

Most commercially available closed-vessel microwave systems are based on multimode microwaves; however, the advantages of high-pressure vessels and focused microwave heating have led to the development of systems that combine both assets. These focused high-pressure, high-temperature microwave extractors consist of an integrated closed vessel and a focused microwave-heated system operating at a very high pressure and temperature.

Dynamic systems for high-pressure microwave extraction were developed much later than open-vessel systems because operating under a high pressure reduces the flexibility afforded by working at atmospheric pressure. In 1999, a microwave-assisted flow digestion system was developed from a novel pressure equilibration system that affords temperatures up to 250°C and pressures in the region of 35 bar with previously slurried solid samples. More recently, Ericsson and Colmsjö developed a dynamic high-pressure microwave-assisted extraction device that allows extraction of solid samples, while avoiding the principal drawback of the previous approach (the relative complexity of the assembly and the inability to introduce unslurried solid samples directly).

Open-Vessel Microwave Systems

The first completely re-engineered laboratory-focused microwave system was introduced by Prolabo in 1986. Most commercial open-vessel microwave systems manufactured since then are of the focused-microwave type, i.e., they use the waveguide as a single-mode cavity. Since their introduction, they have widely been used for sample extraction, substituting in most cases the closed-vessels systems, which as of now are used mainly for carrying out sample digestions.

The principle of focused microwaves is efficient in terms of energy transfer. The electromagnetic energy and the sample are highly efficiently coupled, and a high density is obtained as a result. In fact, according to Prolabo, the coupling efficiency is increased by a factor of 10 compared with the use of a multimode cavity. Moreover, these open systems allow users to add reagents or fresh solvent during the extraction or to connect a device such as a reflux system. In these open devices, a continuously adjustable percentage of the maximum power is used over a given period of time. This is different from a multimode cavity, where pulsed power is used predominantly. One of the main drawbacks of these designs is that they allow the use of only one flask at a time. Automation in some systems can enable sequential use of flasks. One recent development involves the use of four flasks at a time by splitting the microwave energy symmetrically among the flasks at the end of each waveguide.

As noted earlier, not all open-vessel systems are of the focused type. A number of reported applications used a domestic multimode oven to process the samples for analytical purposes, usually with a view to coupling the MAE to some other step of the analytical process (usually the determination step) as described below.

The increased flexibility of open-vessel systems has promoted the design of new microwave-assisted sample treatment units based on focused or multimode (domestic) ovens adapted to the particular purpose. Regarding MAE, examples of these new units include the microwave–ultrasound combined extractor and the focused microwave-assisted Soxhlet extractor.

The microwave–ultrasound combined system has been constructed from a Prolabo Maxidigest 350 single-mode microwave oven, in which a sonotrode is placed at the base for indirect ultrasonic agitation of the sample. The ultrasonic probe is placed at a sufficient distance from the electromagnetic field to avoid interactions and short-circuits. The use of this system has allowed the digestion of Co_3O_4 and olive oil in shorter times than those required by the conventional method (from 3 to 1 h and from 45 to 30 min, respectively).

The focused microwave-assisted Soxhlet extractor design is based on the same principles as a conventional Soxhlet extractor (**Figure 2A**) modified to facilitate accommodation of the sample cartridge compartment in the irradiation zone of a microwave oven. The latter has also been modified by making an orifice at the bottom of the irradiation zone, thus enabling connection of the cartridge compartment to the distillation flask through a siphon as illustrated in **Figure 2B**. The use of this device has provided better results than conventional Soxhlet extraction

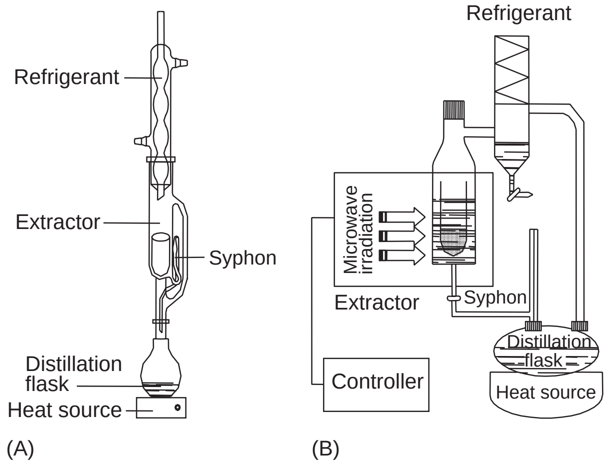

Figure 2 (A) Conventional Soxhlet extractor. (B) Focused microwave-assisted Soxhlet extractor from Prolabo. (Reproduced with permission from Ericsson M and Colmsjö A (2000) Dynamic microwave-assisted extraction. *Journal of Chromatography* 897: 279; © Elsevier.)

in much shorter periods of time. It enables focused microwave-assisted Soxhlet extraction (FMASE), retains the advantages of conventional Soxhlet extraction while overcoming restrictions such as a long extraction time, unsuitability for automation, waste of large volumes of organic solvents, and nonquantitative extraction of strongly retained analytes due to the easier cleavage of analyte–matrix bonds by interactions with focused microwave energy, and, unlike a conventional Soxhlet extractor, allows recycling up to 75–85% of the total extractant volume. In addition, solvent distillation in the FMAS extractor is achieved by electrical heating, which is independent of the extractant polarity, thus avoiding the principal problem of commercial focused microwave devices such as those of the Soxwave series from Prolabo.

Variables Influencing MAE

The performance in microwave-assisted processes depends on a number of variables including the microwave power output, exposure time, solvent used, and sample size used.

Microwave Power and Exposure Time

The principal variables to be considered in developing an MAE method are the applied power and the irradiation (exposure) time. The power and the corresponding time will depend on the type of sample and solvent used. Usually, these two variables have opposing effects: for a given process, the use of high microwave power affords a decreased exposure time; on the other hand, the use of low power entails irradiating the sample for a longer time to apply the same amount of energy.

In theory, one should use a high microwave power to reduce the exposure time as much as possible. Occasionally, however, using a very high power is inadvisable; such as in the case of extraction processes, where complete dissolution of the sample should be avoided in order to minimize the effect of potential interference. In some cases, using a very high microwave power decreases the extraction efficiency through degradation of the sample or its analytes or rapid boiling of the solvent in open-vessel systems (which hinders contact with the sample). Sometimes, the application of microwaves over short, intermittent periods results in substantially increased extraction efficiency relative to the continuous application of microwaves, which causes the solvent to boil and hinders sample–extractant contact.

Temperature and Pressure

Temperature is a key variable in most analytical processes. In MAE, it plays a prominent role and affects the rate of some reactions, the degradation of thermolabile species and the solubilization of some substances, among others. A number of devices have been developed for monitoring or even controlling the temperature.

Pressure is a highly influential factor in closed-vessel systems. The development of vessels capable of withstanding pressures as high as 41 bar has enabled extraction at very high temperatures, thereby increasing dramatically the extraction efficiency and decreasing the exposure time.

Type of Solvent

The choice of solvent in a microwave-assisted sample treatment is a direct function of the type of treatment used. Thus, digestion and hydrolysis are best done with aqueous solutions of pure or mixed acids, whereas selective extraction of organic compounds from a sample matrix usually requires an organic solvent.

MAE of nonpolar compounds usually entails the adoption of a compromise since, although these compounds are more readily dissolved in nonpolar solvents such as n-hexane, the interaction of microwaves with the solvent depends on its dielectric constant (the greater ε is, the stronger the interaction); this requires the use of a mixture of solvents of different polarity in many cases. For example, the most suitable solvent for MAE of organic compounds such as PAHs and PCBs is a mixture of hexane and acetone. In specific cases such as in the extraction of organometal compounds, the use of methanol acidified with acetic acid allows organometals to be extracted in a rapid, efficient manner with no degradation. In some cases such as the extraction of organic compounds of medium or high polarity, pure water can be used as an efficient 'clean' extractant. Although its power of solubility for organic compounds is not as high as those of organic solvents, its interaction with microwaves is much higher due to its high dielectric constant, thus providing high extraction efficiencies and environmentally friendly methods.

Influence of Sample Viscosity and Sample Size on Microwave Heating

The viscosity of a sample reflects its ability to absorb microwave energy because it affects molecular rotation. The effect of viscosity is best illustrated by considering ice. When water is frozen, the water molecules become locked in a crystal lattice. This restricts molecular mobility greatly and makes it difficult for the molecules to align with the microwave field. Thus, the dielectric dissipation factor of ice is low. When the temperature of the water is increased to 27°C, the viscosity decreases and the dissipation factor rises to a much higher value.

The input microwave frequency also affects the penetration depth of the microwave energy. At a given input frequency, the greater the dissipation factor of a sample is, the less it will be penetrated by microwave energy. In large samples with high dissipation factors, the heating that occurs beyond the penetration depth of the microwave energy is due to thermal conductance through molecular collisions. Therefore, temperatures at or near the surface will be higher. Because boiling and other agitation increase the rate of thermal conductance, surface heating is not a problem unless penetration is very low and sample heating is extremely superficial. In such a case, heat losses through the vessel walls can be significant and increase the sample heating time.

Although the small sample size used in most analytical dissolution processes has some advantages, it also has at least one disadvantage: the amount of microwave energy absorbed decreases with decreasing sample size. With small sample sizes, a substantial amount of energy is not absorbed but is reflected. Reflected energy can damage the magnetron, and so, in using small samples for analytical work, it is advisable to employ microwave systems designed to protect the magnetron from reflected power.

Coupling of MAE with Other Operations of the Analytical Process

Online systems using domestic microwave ovens are mainly used for sample digestion. Slurried samples are introduced in a Teflon coil placed in the microwave oven, and after digestion, the solutions are driven to the detector (usually of the atomic type). However, some methods use a domestic microwave oven for extraction purposes, in which the sample is not introduced as a slurry but rather placed in an appropriate vessel that is then introduced into the oven. **Figure 3** shows one of these systems, where the microwave extraction is coupled to subsequent steps

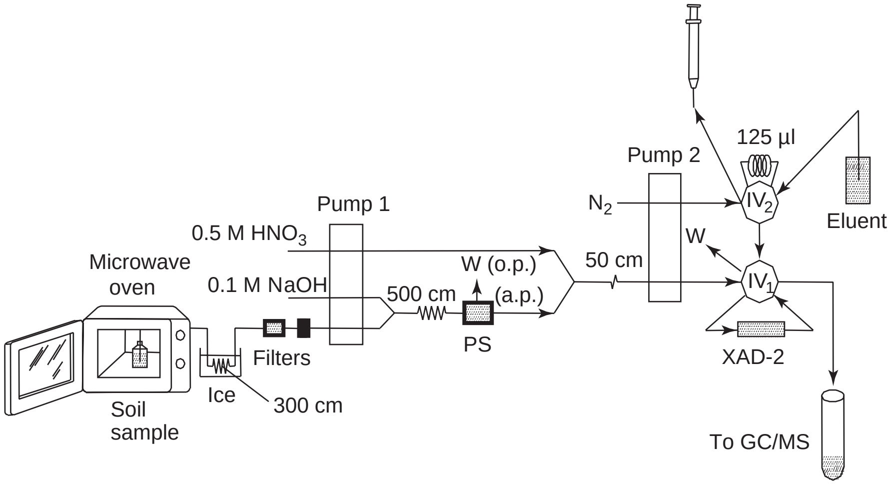

Figure 3 Experimental design for continuous MAE, liquid–liquid extraction, sorption/cleanup of phenol compounds in soil samples. IV, injection valve; PS, membrane phase separator; o.p. and a.p., organic and aqueous phases; W, waste. (Reproduced with permission from Ericsson M and Colmsjö A (2000) Dynamic microwave-assisted extraction. *Journal of Chromatography* 877: 141; © Elsevier.)

via a flow injection (FI) manifold, thus providing partial automation of the analytical process.

Commercial focused microwave devices have been used for coupling the extraction with other steps such as filtration, preconcentration, and/or detection. The experimental setup showed in **Figure 4** uses a piston pump as the interface between the extractor and an FI manifold. Extraction is performed by supplying an appropriate extractant in an automatic manner by means of the piston pump. Then, the sample is irradiated with microwaves for a preset time and the extract aspirated through the other pump channel to a vessel connected to the FI manifold, where the analytes are filtered, preconcentrated, and determined chromatographically in an automatic manner.

The dynamic high-pressure MAE system described previously also allows the extraction to be coupled to a subsequent stage by connecting the extractor outlet to the unit concerned. **Figure 5** illustrates one possible coupling: connecting the extractor to a fluorimetric detector allows the kinetics of the extraction of PAHs from sediments to be monitored.

Applications of MAE

As stated before, MAE has been used mainly for extraction of solid samples. The use of MAE is a continuously expanding area of research at present. As a result, it is difficult to provide a completely updated overview. Rather, selected groups of compounds of interest that have been subjected to MAE are discussed here.

Traditionally, the extraction of PAHs from solid samples has been conducted using a Soxhlet method, which, while being efficient, usually takes more than 6 h and uses large volumes of solvent. Several reports have indicated that PAHs can be extracted from solid matrices such as soil and plant and animal tissues

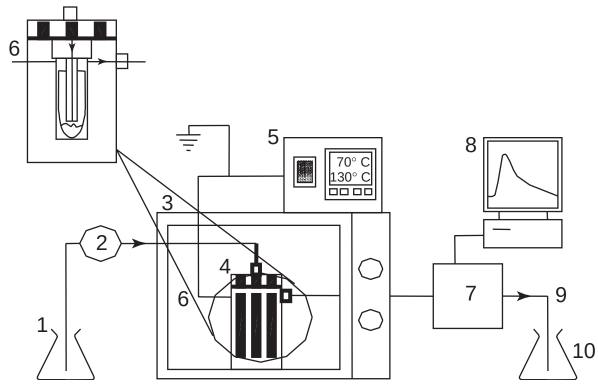

Figure 5 Manifold for dynamic MAE. 1, Solvent; 2, pump; 3, microwave oven; 4, extraction chamber; 5, temperature setpoint controller; 6, thermocouple; 7, fluorescence detector; 8, recording device; 9, restrictor; 10, extractor. (Reproduced with permission from Luque de Castro MD and Luque-García JL (2002) *Acceleration and Automation of Solid Sample Treatment*; © Elsevier.)

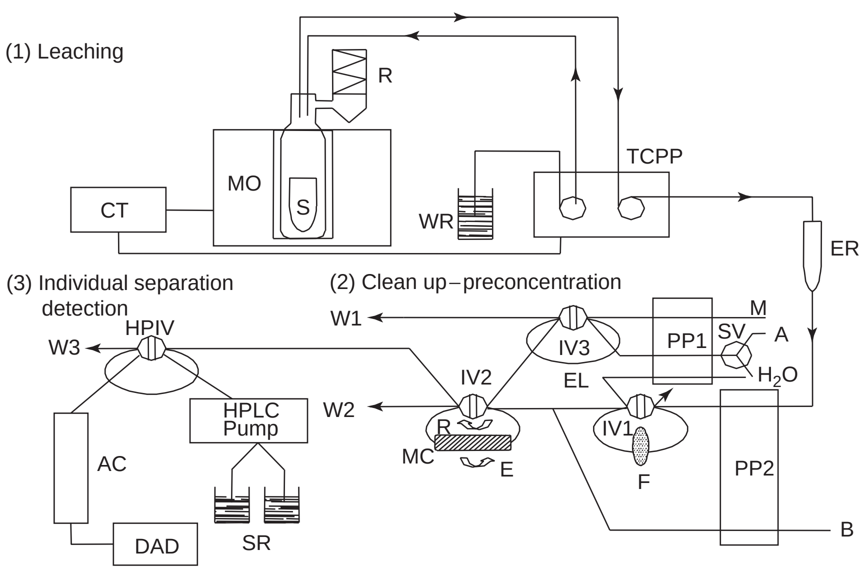

Figure 4 Experimental setup used to integrate MAE with the subsequent steps of the analytical process. (1) Leaching step. CT, controller; MO, microwave oven; S, sample; R, refrigerant; WR, water reservoir; TCPP, two-channel piston pump; ER, extract reservoir; (2) Clean-up/preconcentration step. SV, selecting valve; M, methanol; A, air; B, buffer; PP1 and PP2, peristaltic pumps; F, filter; EL, elution loop; MC, minicolumn; R, retention direction; E, elution direction; IV1–IV3, injection valves; W, waste. (3) Individual separation–detection step. HPIV, high-pressure injection valve; AC, analytical column; DAD, diode array detector; SR, solvent reservoirs. (Reproduced with permission from Luque de Castro MD and Luque-García JL (2002) *Acceleration and Automation of Solid Sample Treatment*; © Elsevier.)

relatively easily using MAE. The most effective solution used for leaching PAHs is an acetone–hexane mixture. The use of water as an extractant has also been tested with a view to developing clean methods using no organic solvents. Because of the high hydrophobicity of PAHs, however, the recoveries were very low in spite of the strong interaction between water and microwaves. One way of using water effectively for MAE of nonpolar compounds (PAHs included) is through micelle formation.

Most studies about MAE of PAHs from solid samples have been conducted using closed-vessel systems and only a few with open-vessel focused microwave devices. Because open-vessel systems operate at atmospheric pressure, the extraction vessel has been used as a reactor in order to perform online purification pretreatments of the total extracts (reagents can be added readily to the medium) or introduce the extracts directly into the determination instrument, as in the focused microwave-assisted extractor with online fluorescent monitoring, which provides a matrix-independent approach to the extraction of PAHs.

MAE has also become a frequent choice for the extraction of PCBs from solid matrices. In fact, this technique has been used to extract PCBs from a wide range of samples including soils, sediments, and animal tissues. Normally, the extractant used is the same as that employed with PAHs; micelle formation have also provided results similar to those obtained using conventional methodologies.

A wide variety of pesticides including organochlorine and organophosphorus compounds, triazines, herbicides, and imidazolinones have been extracted using microwaves, with recoveries $\sim 100\%$ in most cases.

Several MAE methods have also been developed for extraction of organometal compounds such as 3-nitro-4-hydroxyphenylarsonic acid and methylmercury from sediments and animal tissues. The MAE recoveries were consistently higher than those achieved with sonication or Soxhlet extraction. All species recoveries were matrix-dependent, as previously found with other pollutants, but using methanol acidified with acetic acid under optimal MAE conditions afforded rapid, efficient extraction of all analytes.

MAE has also been used as a solid sample treatment prior to speciation analysis, leaving the organometal compound moiety intact. This is a prerequisite for a successful extraction procedure to be applied before speciation analysis can be met by careful optimization of the conditions of the microwave attack. Open-vessel treatment is preferred to the pressurized bomb systems commonly used in analysis of total metals because it offers milder reaction conditions (the increase in temperature is governed to a great extent by the boiling point of the solvent) and easier control of process variables.

Other organic compounds such as phenols, hydrocarbons, polymer additives, and natural compounds have also been extracted from a wide range of samples with good results.

One of the most interesting fields of application of MAE in food analysis is the extraction of lipids. This step, traditionally performed with conventional Soxhlet extraction, has been performed with the focused microwave-assisted Soxhlet extractor prototype of **Figure 2B**. Extraction of oil from olives, sunflower seeds, and soyabeans; extraction of the lipid fraction of dairy products (milk and cheese); and extraction of fatty acids from precooked and sausage foods have significant advantages over conventional methods, including dramatically reduced extraction times, lower degradation of thermolabile analytes, and acceleration of other analytical steps such as hydrolysis in milk samples, in addition to completeness of analyte extraction, which is not always achieved with conventional methods.

The extraction of metals and metalloids has also been assisted by microwaves, most often for speciation analysis. Metals such as Fe, Zn, Cu, As, Se, Hg, Pb, Co, Cr, Ni, and Cd have been extracted using acid solutions (usually HNO_3) in a focused open-vessel system for periods ranging from 3 min for soil samples to 50 min for coal (where the metals are much more strongly retained). MAE has been used both as a pretreatment for chemical speciation (mainly to discriminate As and Se species) and as a means of accelerating fractionation of metals in sewage sludge for operation and/or functional speciation. The former type of application has usually been developed using open-vessel systems coupled to various types of equipment for individual separation and/or determination (e.g., hydride generation–atomic fluorescence detection, electrochemical detection, and liquid chromatography–hydride generation–inductively coupled plasma–mass spectrometry) following extraction.

See also: **Countercurrent Chromatography**: Solvent Extraction with a Helical Column. **Extraction**: Solvent Extraction Principles; Solvent Extraction: Multistage Countercurrent Distribution; Pressurized Fluid Extraction; Supercritical Fluid Extraction.

Further Reading

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Pressurized Fluid Extraction

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Introduction

Over the past few years, a technique based on the use of solvents at a high pressure and high temperature without reaching the critical point has gradually emerged as an efficient means for increasing automatability, shortening process times, and reducing the amount of solvent required to digest or leach solid samples. Such features as the low consumption of organic solvents and the expeditiousness of determinations make this technique especially suitable for environmental analysis, the area where it has expanded to the greatest extent.

The principal aspects of the technique, in its two modes (static and dynamic), the devices typically employed by each, and their amenability to coupling with subsequent operations of the analytical process, which is one of the most interesting aspects of this technique, are discussed in this article. The main variables affecting the extraction process as well as the applications are also commented on.

Operational Pressurized Fluid Extraction Modes

Depending on the way the sample and extractant are brought into contact, pressurized fluid extraction (PFE) can be implemented in three different operational modes, static, dynamic, and static–dynamic.

In the static extraction mode, the sample is soaked in the extractant and the liquid flow is halted over a preset interval, after which it is propelled to the collection vessel, usually by a nitrogen stream. This extraction mode allows penetration of the extractant, and it is especially useful when the analytes cannot be removed expeditiously from the matrix.

In the dynamic extraction mode, the extractant is pumped through the sample at a preset flow rate. This mode allows the analyte to be exposed continuously to the pure (clean) solvent, thus favoring displacement of the analyte's partitioning equilibrium to the extractant.

Which extraction mode is better remains a controversial issue. The static mode provides longer contact between the sample and solvent, swells the matrix, and facilitates penetration of the extractant in its interstices. However, it is not suitable when the partitioning equilibrium is displaced slightly to the solubilization of the analytes in the extractant. The dynamic mode has the advantage that it puts the sample into contact with fresh extractant continuously but requires a larger amount of extractant and causes dilution of the extracts, which usually calls for a preconcentration step prior to determination of the analytes. The static–dynamic mode, which consists of a combination of the two previous modes, can be used to enjoy the advantages of both modes while circumventing their shortcomings.

Pressurized Fluid Extractors

PFE has been carried out using both commercial and laboratory-built extractors, depending mainly on the operational mode used. While static PFE is usually carried out using commercial equipment (e.g., Dionex accelerated solvent extraction (ASE) 200 and ASE 300 or, more recently, the one commercialized by Applied Separation), the dynamic mode has been implemented only in laboratory-built extractors as nowadays there is no commercially available equipment. **Figure 1A** and **1B** shows schematic diagrams of a static and dynamic pressurized fluid (PF) extractor, respectively. The former (**Figure 1A**) consists of the following parts: (1) a reservoir for storing the liquid extractant; (2) a high-pressure pump for propelling the extractant through the system; (3) an extraction chamber, consisting of a